MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION (2024 ADMISION ONWARDS)

MG5DSCMAT300: A First Course in Complex Analysis

Duration: 2 hrs **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to en hance their overall outcome attainability.

Part A

Answer any five questions. Each question carries 2 marks

1. Evaluate $\int_0^1 (1+it)^2 dt$ [U][1]
2. State the Cauchy –Goursat theorem. [K][1]
3. Find the Maclaurian series for $f(z)$=$z$sin$z$. [U][6]
4. Evaluate $\lim_{n\to\infty} \frac{n^2+in}{n^2+1}$. [U][2]
5. Find the singular points, if any, of the function $\frac{sinz^2}{z^2+1}$ [U][2]
6. State the formula for finding the residue of a function f(z) at a pole of order m. [U][2]
7. Define improper integral of a continuous function f(x) over the interval $-\infty < x < \infty$. [K][7]
8. State Jordan's inequality. [K][7]

Part B Answer any five questions. Each question carries 6 marks

9. State and prove the Principle of Deformation of Paths [K][2]
10. Evaluate $\int_C \frac{cosz}{z(z^2+8)} dz\ where$ C is the circle $|z| = 3$. [A][3]
11. Using the formal definition of limit, prove that the sequence z_n= in/(n+1) converges to i. [U][2]
12. Find Taylor series of $f(z)$= 1/ z^2 about z=2 and specify the region in which series is valid. [A][6]
13. Find the residue at z = 0 of the function cotz /z^4 . [U][2]
14. Determine the nature of singularity of f(z) $\frac{1}{z(e^z-1)}$. [An][5]
15. Evaluate $\int_0^{2\pi} \frac{d\theta}{3+2sin\theta}$. [E][7]
16. State Jordan;s lemma [K][7]

Part C Answer any three questions. Each question carries 10 marks

17. State and prove Liouville's theorem. [An][4]
18. (a) State Taylor's theorem.

    (b) Show that $\frac{1+2z^2}{z^3+z^5} = \frac{1}{z^3} + \frac{1}{z} - z + z^3 - z^5 \ldots \ (0< |z| < 1)$ [U][6]
19. Find Laurent series expansion for $f(z)= z^2\sin(1/z^2)$ and $f(z) = z^3\cos(1/z^3)$. [C][6]
20. Describe the different types of isolated singularities with examples. [U][2]
21. Use residues to find the Cauchy Principal value of the integral $\int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2}$. [E][7]
22. Show that $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}$, a $>|b|$. [E][7]

****